Notes: Eisenbach and Phelan (RFS, 2026)

Fragility of Safe Asset Markets

Contents

1 Big picture

- **Question.** Why can markets for safe assets, especially U.S. Treasuries, suddenly become fragile and suffer price crashes during stress? Why did March 2020 feature a "dash-for-cash" rather than the usual flight-to-safety rally?
- **Motivation.** Safe assets normally appreciate when risky assets fall. In March 2020, Treasury prices initially rose as stocks fell, but then Treasury notes and bonds also sold off sharply. Foreign official investors and mutual funds were large sellers, and part of their selling appears preemptive rather than mechanically forced by immediate cash needs.
- **Core environment.** The model has three relevant forces: liquidity investors who may need cash and hold safe assets as liquidity insurance; safety investors who demand safe assets when risky assets become unattractive; and dealers who intermediate trades but face balance sheet costs.
- **Main mechanism.** Liquidity investors who are not currently hit by a liquidity shock decide whether to sell now or wait. If many other investors sell now, dealers accumulate inventory. That inventory depresses the future price conditional on a future liquidity shock. Hence other investors' current sales can make it privately optimal to sell preemptively today.
- **Strategic complementarity.** A run is possible when selling today raises the payoff from selling today relative to waiting. The complementarity is not automatic. It requires either sufficiently elastic dealer balance sheet costs, investor risk aversion, or sequential trade execution in which investors prefer to sell before others in an over-the-counter market.
- **Global-game result.** With a small amount of noise about aggregate liquidity risk, the multiple-equilibrium complete-information game becomes a unique threshold equilibrium. Below a threshold s^*, strategic liquidity investors hold; above it, they all sell. The threshold is the paper's measure of market stability.
- **Surprising result.** Flight-to-safety demand can be destabilizing. More demand from safety investors raises the current price, which encourages liquidity investors to sell now. It also lowers dealer inventory and raises the future price, which discourages selling now. When the market is already fragile, the first effect can dominate and safety demand can trigger a dash-for-cash.
- **Policy implication.** The relevant policy question is not only how many assets the central bank buys, but how the policy changes beliefs about future market liquidity. A credible future purchase facility can stabilize the market by improving the expected future sale price. Relaxing dealer constraints can help, but a short-run-only relaxation can backfire if it makes selling immediately more attractive.
- **Takeaway.** Safe asset markets can be deep in normal times and still fragile in stress. The fragility comes from the interaction of liquidity needs, dealer balance sheet constraints, and strategic preemptive selling.

2 Model environment and key objects

2.1 Agents, assets, and timing

The economy has two trading dates, $t = 0, 1$. The central asset is a safe asset with fundamental value normalized to one. It is safe in the sense of payoff risk, but it is not perfectly liquid in the sense that large imbalances must be absorbed by dealers with balance sheet costs.

There are three classes of agents:

- **Liquidity investors.** They initially hold the safe asset as protection against liquidity shocks. If hit by a shock, they must sell immediately to obtain cash.
- **Dealers.** They intermediate safe-asset sales. Dealers are competitive and risk neutral, but taking inventory is costly because it uses scarce balance-sheet space.
- **Safety investors.** They are risk averse investors who rebalance toward safe assets when risky assets become less attractive. They generate flight-to-safety demand.

The baseline analysis first studies liquidity investors and dealers. Safety investors are added later to study the interaction between flight-to-safety and dash-for-cash.

2.2 Liquidity investors and the decision to sell

Each liquidity investor starts with one unit of the safe asset. At date 0, a fraction $s_0 \in (0, 1)$ receives a liquidity shock and must sell at price p_0. The remaining fraction $1 - s_0$ is strategic: these investors choose whether to sell at date 0 even though they do not currently need cash.

If a strategic investor waits, she may be hit by a date-1 liquidity shock with probability $s_1 \in (0, 1)$ and then sell at price p_1^s. If she is not hit by a date-1 shock, she keeps the safe asset and obtains continuation value $v > 1$. The value v captures the convenience yield or liquidity service from holding the safe asset.

The strategic investor sells preemptively at date 0 if and only if

$$p_0 > s_1 p_1^s + (1 - s_1)v. \tag{1}$$

Let $\lambda \in [0, 1]$ be the fraction of strategic liquidity investors who sell at date 0. The payoff gain from selling now rather than waiting is

$$\pi(\lambda) = p_0(\lambda) - s_1 p_1^s(\lambda) - (1 - s_1)v. \tag{2}$$

The complete-information game has three natural possibilities:

- **Hold equilibrium:** if $\pi(0) < 0$, no strategic investor sells.
- **Run equilibrium:** if $\pi(1) > 0$, all strategic investors sell.
- **Mixed equilibrium:** if $\pi(\lambda^*) = 0$ for some interior λ^*, strategic investors are indifferent.

The convenience yield $v > 1$ is important because it allows a normal-times hold equilibrium. Without $v > 1$, safe-asset holders would have too little private benefit from waiting, and preemptive selling would be much harder to rule out.

2.3 Dealer balance sheet costs and safe-asset prices

Dealers begin with zero inventory. If they absorb quantity q, they pay a strictly convex balance sheet cost $C(q)$, with $C(0) = 0$ and $C(1) < 1$. Competitive dealer pricing implies zero profits.

At date 0, if dealers buy quantity q_0, the date-0 zero-profit condition is

$$(1 - p_0)q_0 - C(q_0) = 0. \tag{3}$$

At date 1, if dealers already carry inventory q_0 and buy additional quantity q_1^s, the zero-profit condition is

$$(1 - p_0)q_0 + (1 - p_1^s)q_1^s - C(q_0 + q_1^s) = 0. \tag{4}$$

Solving these equations gives the key price formulas:

$$p_0(q_0) = 1 - \frac{C(q_0)}{q_0}, \qquad p_1^s(q_0, q_1^s) = 1 - \frac{C(q_0 + q_1^s) - C(q_0)}{q_1^s}. \tag{5}$$

The price at date 1 is lower than the price at date 0 because the additional date-1 purchase comes on top of existing inventory. Convex balance sheet costs make the future sale price especially sensitive to current inventory.

Sales are determined by liquidity shocks and strategic preemptive selling:

$$q_0 = s_0 + (1 - s_0)\lambda, \tag{6}$$
$$q_1^s = s_1(1 - s_0)(1 - \lambda). \tag{7}$$

Thus λ affects both dates. More date-0 strategic selling directly lowers p_0 and indirectly lowers p_1^s by raising dealer inventory.

2.4 Safety investors and flight-to-safety demand

Safety investors are introduced to capture the usual increase in demand for safe assets during stress. They have utility

$$u(c_0, w) = c_0 + w - \frac{1}{2}\kappa w^2, \tag{8}$$

where w is future wealth and $\kappa > 0$ captures curvature.

They hold a risky asset with payoff z, mean μ_z, and variance σ_z^2, plus the safe asset. In equilibrium, their demand for the safe asset is linear:

$$q_0^S = a - bp_0, \qquad a = \frac{1}{\kappa} - \mu_z Z, \qquad b = \frac{1}{\kappa}. \tag{9}$$

A negative shock to the expected payoff of the risky asset lowers μ_z, raises a, and shifts out demand for the safe asset. This is the model's flight-to-safety shock.

3 Models and theoretical structure

3.1 Strategic complementarities in the liquidity-investor game

Using the dealer price functions, the payoff gain from selling preemptively is

$$\pi(\lambda, s_0, s_1) = p_0(\lambda, s_0) - s_1 p_1^s(\lambda, s_0, s_1) - (1 - s_1)v. \tag{10}$$

Strategic complementarities arise when

$$\frac{\partial \pi}{\partial \lambda} > 0. \tag{11}$$

The economic meaning is simple: if more other strategic investors sell, my incentive to sell also rises. This is the formal condition for a dash-for-cash mechanism.

There are two forces when λ rises:

- **Current-price effect.** More date-0 sales reduce p_0. This makes selling now less attractive, so it is stabilizing.
- **Inventory/future-price effect.** More date-0 sales increase dealer inventory and reduce p_1^s. This makes waiting more dangerous, so it is destabilizing.

The destabilizing inventory effect must dominate the stabilizing current-price effect. A sufficient condition is given in terms of the elasticity of marginal balance sheet costs,

$$\eta \equiv \frac{qC''(q)}{C'(q)}. \tag{12}$$

With a general convex cost function, strategic complementarities arise uniformly when date-1 liquidity risk and marginal-cost elasticity are jointly high:

$$s_1 \eta > 2. \tag{13}$$

Interpretation: future liquidity risk makes the future price relevant, and elastic balance sheet costs make future prices very sensitive to the dealer inventory inherited from date 0.

3.2 Parametric balance sheet costs

The paper studies cost functions of the form

$$C(q) = cq^n, \qquad c > 0, \quad n > 1. \tag{14}$$

Then $\eta = n - 1$. The curvature parameter n governs how quickly dealer marginal costs rise as inventory grows.

Important cases:

- **Quadratic costs,** $n = 2$. Strategic complementarities cannot arise in the pooled-execution, risk-neutral baseline.

- **Cubic costs,** $n = 3$. Strategic complementarities arise when future liquidity risk exceeds

$$\widetilde{s} = 1 - \sqrt{\frac{1}{3}} \approx 0.42. \tag{15}$$

- **Higher powers,** $n > 3$. The threshold for strategic complementarities falls as n rises. More elastic balance sheet costs make runs easier to generate.

This matters because the paper does not assume that safe assets are fragile by construction. The fragility appears only when the dealer sector's capacity becomes sufficiently nonlinear or when other realistic frictions strengthen the desire to sell early.

3.3 Risk aversion and sequential execution

The paper then shows that two realistic extensions can generate strategic complementarities even with less elastic balance sheet costs.

First, if liquidity investors are risk averse, the bad future state becomes more painful. With quadratic costs and constant relative risk aversion utility

$$u(x) = \frac{x^{1-\gamma}}{1-\gamma}, \qquad \gamma > 0, \tag{16}$$

strategic complementarities arise when liquidity risk and/or risk aversion are sufficiently high. The paper states the sufficient condition as

$$s_1(2 - s_1) > (1 - cs_1)^{\gamma}. \tag{17}$$

Second, in an over-the-counter market, trades are often executed sequentially rather than pooled. If the expected position in the queue is half of total order flow, the expected prices under quadratic costs become

$$\mathbb{E}[p_0(q_0)] = 1 - \frac{1}{2}cq_0, \tag{18}$$

$$\mathbb{E}[p_1^s(q_0, q_1^s)] = 1 - 2cq_0 - \frac{1}{2}cq_1^s. \tag{19}$$

Sequential execution strengthens the date-0 inventory effect on the future price. An investor who sells today expects to front-run part of today's selling, but if she waits until date 1 she bears the full inventory effect left by date-0 sales.

With quadratic costs and sequential execution, the payoff gain is

$$\pi(\lambda, s_0, s_1) = 1 - \frac{c}{2}[s_0 + (1 - s_0)\lambda] \tag{20}$$

$$- s_1 \left(1 - 2c[s_0 + (1 - s_0)\lambda] - \frac{c}{2}s_1(1 - s_0)(1 - \lambda)\right) \tag{21}$$

$$- (1 - s_1)v. \tag{22}$$

Strategic complementarities arise when

$$s_1 > \widetilde{s} = 2 - \sqrt{3} \approx 0.27. \tag{23}$$

This result is important because it shows that sequential trade execution can make fragility arise even under the simple quadratic cost specification.

3.4 Complete information and global-game equilibrium

For tractability, the paper then sets $s_0 = s_1 = s$ and uses the quadratic-cost, sequential-execution case. The payoff gain becomes

$$\pi(\lambda, s) = \frac{c}{2}\Big[(4s-1)\big(s+(1-s)\lambda\big) + s^2(1-s)(1-\lambda)\Big] - (1-s)(v-1). \tag{24}$$

For low s, $\pi(\lambda, s) < 0$: no strategic liquidity investor sells. For high s, $\pi(\lambda, s) > 0$: all strategic investors sell. For intermediate s, strategic complementarities can create multiple equilibria.

To select a unique equilibrium, the paper introduces noisy private signals. Each investor observes

$$\widehat{s}_i = s + \sigma_\varepsilon \varepsilon_i, \tag{25}$$

where the noise is arbitrarily small. The global-game equilibrium is a threshold equilibrium. In the limit $\sigma_\varepsilon \to 0$, the switching threshold s^* solves

$$\int_0^1 \pi(\lambda, s^*)\,\mathrm{d}\lambda = 0. \tag{26}$$

The equilibrium behavior is:

$$\begin{cases} s < s^*: & \text{strategic liquidity investors hold; only forced sellers sell,} \\ s > s^*: & \text{strategic liquidity investors run; all sell at date 0.} \end{cases} \tag{27}$$

The threshold s^* is therefore a measure of market stability. A higher s^* means the market can withstand more liquidity risk before running.

3.5 Equilibrium price crash

In the threshold equilibrium, date-0 supply jumps from s to 1 when s crosses s^*. Under the simplified price function, the equilibrium date-0 price is

$$p_0^*(s) = \begin{cases} 1 - cs, & s < s^*, \\ 1 - c, & s > s^*. \end{cases} \tag{28}$$

This is the formal representation of a dash-for-cash. The underlying liquidity risk s changes continuously, but the equilibrium price can move discontinuously because behavior switches from holding to running.

Dealer balance sheet costs reduce stability:

$$\frac{\partial s^*}{\partial c} < 0. \tag{29}$$

The liquidity convenience value raises stability:

$$\frac{\partial s^*}{\partial v} > 0. \tag{30}$$

Thus tighter dealer capacity, interpreted partly as balance sheet regulation such as the Supplementary Leverage Ratio, can increase the fragility of safe-asset markets.

3.6 Welfare and a Pigouvian tax

Selling without a genuine liquidity need is inefficient because liquidity investors value holding the safe asset at $v > 1$, while dealers value the asset at its fundamental value and face inventory costs.

Ex post social welfare is

$$W = v - (q_0 + q_1)(v - 1) - C(q_0 + q_1). \tag{31}$$

Total sales can be decomposed as

$$q_0 + q_1 = \left(2s - s^2\right) + (1 - s)^2\lambda. \tag{32}$$

The first term is genuine liquidity-driven sales. The second term is unnecessary preemptive selling. Welfare is therefore maximized at $\lambda = 0$ for a given s, and ex ante welfare rises when the threshold s^* rises.

The paper considers a Pigouvian tax τ on safe-asset sales, rebated to investors who held the asset at the beginning of the period. Since the rebate does not depend on the individual sale decision, the payoff gain becomes

$$\pi = (1 - \tau)p_0 - s(1 - \tau)p_1^s - (1 - s)v. \tag{33}$$

Because $p_1^s < p_0$, the tax reduces the incentive to sell preemptively. A sufficiently high tax can eliminate runs, though it may create undesirable transfers from investors with genuine liquidity needs to other investors. A more targeted state-contingent version resembles swing pricing: it becomes active only when aggregate sales exceed a threshold.

3.7 Safety investors and the ambiguous effect of flight-to-safety

Combining dealer demand with safety investor demand gives the date-0 price

$$p_0(q_0) = \frac{1 + ac}{1 + bc} - \frac{c}{1 + bc} q_0. \tag{34}$$

At date 1, safety investors are inactive in the baseline extension, but their date-0 purchases reduce dealer inventory. The date-1 price becomes

$$p_1^s(q_0, q_1^s) = 1 - 2c\frac{q_0 + b - a}{1 + bc} - cq_1^s. \tag{35}$$

An increase in safety demand a has two effects:

- It raises p_0, making it more attractive for liquidity investors to sell now. This is destabilizing.
- It reduces dealer inventory and raises p_1^s, making it less costly to wait. This is stabilizing.

The paper's key result is

$$\frac{\partial \pi}{\partial a} > 0 \quad \Longleftrightarrow \quad s < \frac{1}{2}. \tag{36}$$

Thus flight-to-safety demand increases the incentive to sell when liquidity risk is low. At first this sounds counterintuitive, but it follows from the fact that when s is low, the future sale price matters less because the investor is unlikely to need to sell at date 1. The higher date-0 price dominates.

In global-game terms:

$$\frac{\mathrm{d}s^*}{\mathrm{d}a} > 0 \quad \Longleftrightarrow \quad s^* > \frac{1}{2}. \tag{37}$$

A stable market becomes more stable when safety demand rises. A fragile market becomes more fragile. This is the paper's main explanation for why a normal flight-to-safety can coexist with, and even trigger, a dash-for-cash.

3.8 Correlated liquidity and safety shocks

In stress episodes, liquidity risk s and safety demand a may rise together. The equilibrium date-0 price with safety investors is

$$p_0^*(s, a) = \begin{cases} \dfrac{1}{1 + bc}\left[1 - c(s - a)\right], & s < s^*(a), \\ \dfrac{1}{1 + bc}\left[1 - c(1 - a)\right], & s > s^*(a). \end{cases} \tag{38}$$

In the hold region, equal-sized increases in liquidity risk and safety demand offset each other in prices. If safety demand rises more than liquidity risk, the safe asset rallies. If liquidity risk catches up and crosses the threshold, the market can switch to the run region and prices crash.

This maps directly to March 2020: early in the episode, Treasury prices rose as risky asset prices fell; then, in mid-March, liquidity pressure and strategic sales dominated, and Treasury prices fell together with stocks.

4 Mechanism, credibility, and policy logic

4.1 Why the mechanism is not simply "Treasuries became risky"

The model does not assume that the safe asset has payoff risk. The fundamental value remains one. Fragility arises because of market liquidity and intermediation capacity, not default risk. The same asset can therefore be fundamentally safe and still experience a price crash when many investors try to transform it into cash at the same time.

This distinction is central for interpreting March 2020. Treasury notes and bonds were still safe in payoff terms, but they were not perfect cash substitutes in the relevant stress state. Cash and very short Treasury bills were closer to the consumption/liquidity good in the model.

4.2 What makes the model produce regime shifts

The paper needs three ingredients to generate realistic regime shifts:

- **A normal-times hold equilibrium.** The convenience value $v > 1$ makes it privately valuable to keep the safe asset in normal times.
- **A capacity-constrained intermediary sector.** Convex dealer costs make future prices sensitive to current inventory.
- **Strategic uncertainty.** With global-game noise, investors do not simply coordinate on an arbitrary equilibrium; the model selects a threshold where behavior switches.

Together, these ingredients explain why safe-asset markets can look deep and resilient most of the time, but fail abruptly once the state variable crosses a critical region.

4.3 How the model interprets March 2020

The paper reads March 2020 as a perfect storm:

- Post-crisis regulation and balance sheet constraints made dealer intermediation more costly.
- The pandemic shock created genuine cash needs for some investors.
- The decline in risky assets generated strong flight-to-safety demand from other investors.
- In a fragile market, the flight-to-safety itself could make selling now more attractive for liquidity investors, helping trigger a dash-for-cash.

A key empirical motivation is that some observed Treasury sales appear preemptive. Foreign official agencies sold Treasury bonds but reduced total dollar assets much less; mutual funds sold more Treasuries than needed to meet outflows. This supports the paper's focus on strategic selling rather than only mechanical liquidity needs.

4.4 Policy intervention: asset purchases

A purchase facility announced at date 0 and implemented at date 1 directly improves the date-1 expected sale price. If the facility buys quantity q_1^F at date 1, the payoff gain falls with q_1^F:

$$\frac{\partial \pi}{\partial q_1^F} = -sc < 0. \tag{39}$$

The stabilizing effect is stronger when liquidity risk s is high and dealer balance sheet costs c are high. The intuition is that a future backstop reduces the fear of being forced to sell into a bad future market.

The announcement can matter more than the purchases themselves because it changes the selected equilibrium. If the announcement raises the threshold from s^*_{pre} to s^*_{post}, then states between these thresholds move from the run equilibrium to the hold equilibrium, causing a discrete price improvement.

4.5 Policy intervention: dealer constraints

Dealer constraints are central in the model. Reducing them can increase market stability, but timing matters. A relaxation that mainly improves date-0 selling conditions can make current selling more attractive and may worsen preemptive sales. A relaxation that credibly improves future intermediation conditions lowers the incentive to run.

The paper uses this logic to discuss the March 2020 policy sequence. Large repo operations did not fully stabilize Treasuries because they did not necessarily relax the binding SLR-related balance sheet constraint. Outright purchases and later temporary SLR exemptions were more directly connected to easing the constraint that mattered for dealer intermediation.

4.6 Limitations and interpretation

The model is stylized. It treats liquidity shocks as exogenous, abstracts from the exact maturity boundary between cash-like bills and longer Treasuries, and focuses on a simplified dealer sector. Hedge fund basis-trade unwinds, margin spirals, and other institutional details can also contribute to Treasury-market stress.

The value of the model is not that it explains every trade in March 2020. Its contribution is a clean mechanism for why preemptive sales can arise in safe asset markets and why policies that affect future liquidity can have large, discontinuous effects.

5 Figures, propositions, and empirical messages

5.1 Figure-by-figure reading

5.1.1 Figure 1 The market for U.S. Treasuries in early 2020

Read:

In early 2020, Treasuries first behaved like safe assets, rising as stocks fell, but then Treasury prices fell together with stocks. Dealer balance sheets came under pressure, and foreign investors plus mutual funds sold unusually large amounts. **Message:** The episode looks like a regime shift, not a smooth movement in fundamentals.

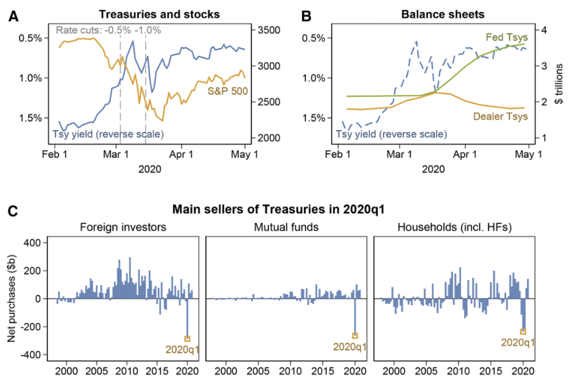

Figure 1
The market for U.S. Treasuries in early 2020
Market yield on 10-year Treasuries, Federal Reserve rate cuts, and S&P 500 index (panel A). Treasuries on the balance sheets of the Federal Reserve (outright holdings) and of Primary Dealers (net positions and gross reverse repo, panel B). Net purchases of Treasuries (panel C). For details on the data, see Appendix A.

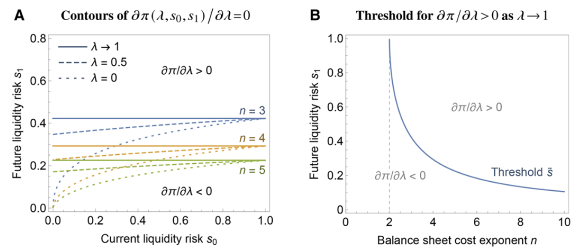

Figure 2
Transition from strategic substitutes to strategic complements
Panel A shows contour plots of $\partial\pi(\lambda, s_0, s_1)/\partial\lambda = 0$ for different values of λ as well as different exponents n in the balance sheet cost function $C(q) = cq^n$ (with $c = 0.25$). Panel B shows the threshold $\tilde{s} = 1 - \sqrt{(n-2)/n}$.

5.1.2 Figure 2 Transition from strategic substitutes to strategic complements

Read:

The boundary between strategic substitutes and complements shifts as the power n in $C(q) = cq^n$ changes. **Message:** More elastic dealer marginal costs make strategic complementarities easier to obtain.

5.1.3 Figure 3 Incentive to sell and equilibria

Read:

The payoff gain $\pi(\lambda, s)$ changes with liquidity risk s. Low s gives a hold equilibrium; high s gives a run; intermediate s can have multiple equilibria. **Message:** Fragility appears when liquidity risk is high enough that others' selling raises my incentive to sell.

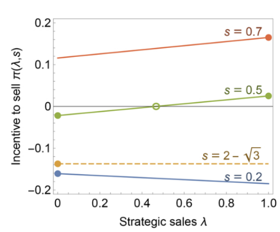

Figure 3
Incentive to sell and equilibria
The figure shows the payoff gain π for different values of liquidity risk s. Circles represent equilibria of the game under complete information. Parameters: $c = 0.25$, $v = 1.2$.

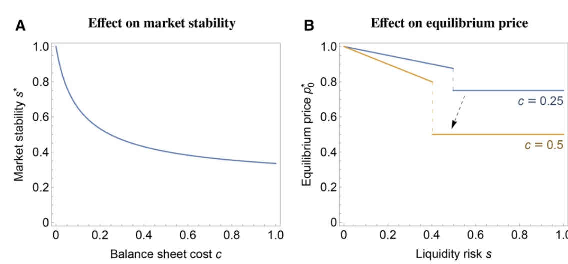

Figure 4
Effect of balance sheet costs on market stability and equilibrium price
Panel A shows market stability measured by the equilibrium threshold s^* as a function of the dealer balance sheet cost c. Panel B shows the equilibrium price at date 0 p_0^* as a function of liquidity risk s for different values of dealer balance sheet cost c. Parameter: $v = 1.2$.

5.1.4 Figure 4 Effect of balance sheet costs on market stability and equilibrium price

Read:

Higher dealer balance sheet costs lower the threshold s^* and create a larger price drop at the run threshold. **Message:** Dealer constraints make the safe-asset market more fragile.

5.1.5 Figure 5 Effect of flight-to-safety on equilibrium market stability

Read:

More safety demand raises stability in a stable market but lowers stability in a fragile market. **Message:** Flight-to-safety is stabilizing only when the market is sufficiently resilient.

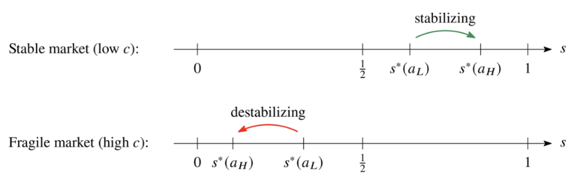

Figure 5
Effect of flight-to-safety on equilibrium market stability
The figure shows the effect of an increase in safety investor demand from a_L to a_H on market stability s^* for different levels of dealer balance sheet cost c.

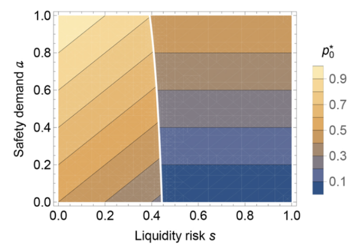

Figure 6
Equilibrium price for combinations of liquidity and safety shocks
The figure shows a contour plot of the equilibrium price p_0^* as a function of liquidity risk s and safety demand a. Parameters: $v = 1.2$, $b = 1$, $c = 1$.

5.1.6 Figure 6 Equilibrium price for combinations of liquidity and safety shocks

Read:

The price surface in (s, a) space shows a cliff. A rise in safety demand can offset liquidity risk in the hold region, but a threshold crossing creates a sudden crash. **Message:** The model can generate both the initial Treasury rally and the later dash-for-cash.

5.1.7 Figure 7 Federal Reserve purchases and foreign official sales of Treasuries

Read:

Fed purchases, foreign official sales, and Treasury yields line up with a transition from stress to stabilization once purchases became larger and more credible. **Message:** Policy credibility about future liquidity provision can switch the market back to the hold equilibrium.

5.1.8 Figure 8 Effects of date-1 purchase facility announced at date 0

Read:

A date-1 purchase facility increases the stability threshold and can create an immediate date-0 price jump upon announcement. **Message:** Announcement effects are natural in a regime-shift model.

5.1.9 Figure 9 Federal Reserve interventions in Treasury markets

Read:

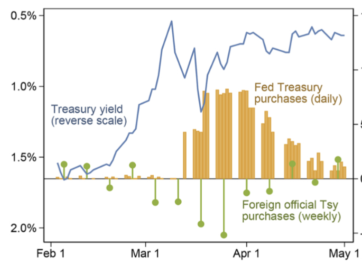

Figure 7
Federal Reserve purchases and foreign official sales of Treasuries
The figure shows the market yield on 10-year Treasuries as well as Federal Reserve Treasury purchases and foreign official agencies' net Treasury purchases inferred from changes in custody holdings. For details on the data, see Appendix A.

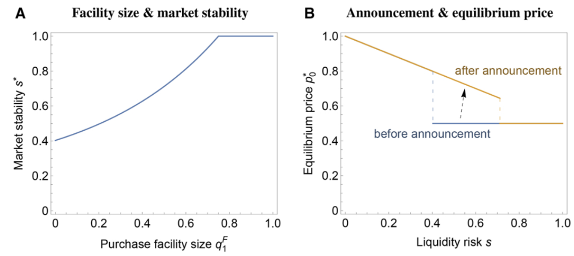

Figure 8
Effects of date-1 purchase facility announced at date 0
Panel A shows the effect of the facility size q_1^F on date-0 market stability as measured by the equilibrium threshold s^*. Panel B shows the effect of the announcement of a facility with $q_1^F = 0.5$ on the equilibrium price at date 0. Parameters: $\upsilon = 1.2$, $c = 0.5$.

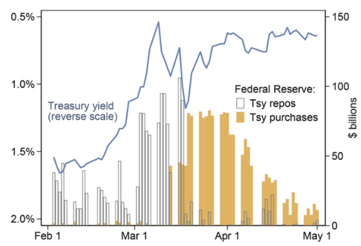

Figure 9
Federal Reserve interventions in Treasury markets
The figure shows the market yield on 10-year Treasuries and two types of Federal Reserve interventions: Treasury repos (lending against Treasury securities) and outright Treasury purchases. For details on the data, see Appendix A.

Repo operations expanded before outright purchases, but price recovery coincided more closely with purchases and later SLR relief. **Message:** Policy must target the binding constraint; funding support is not enough if balance-sheet capacity is binding.

5.2 Proposition-by-proposition reading

Result	Economic content
Lemma 1	Dealer zero-profit pricing gives $p_0(q_0)$ and $p_1^s(q_0, q_1^s)$. The future price is lower because new sales arrive on top of old inventory.
Corollaries 1–2	Prices fall with asset sales and dealer inventory. Date-0 sales depress the date-1 price through the inventory channel.
Proposition 1	Strategic complementarities arise when future liquidity risk and the elasticity of marginal balance sheet costs are high enough.
Proposition 2	With power costs $C(q) = cq^n$, quadratic costs cannot generate complementarities in the pooled baseline, but cubic and higher costs can.
Proposition 3	Investor risk aversion makes waiting more painful in the bad future state, helping generate complementarities.
Proposition 4	Sequential execution strengthens the incentive to sell early and can generate complementarities with quadratic costs.
Proposition 5	Global-game noise selects a unique threshold equilibrium. The threshold s^* is the central market-stability statistic.
Corollary 5	Higher dealer balance sheet costs reduce stability; higher continuation value of holding the safe asset increases stability.
Proposition 6	A Pigouvian tax on sales reduces the incentive to run and can eliminate runs if sufficiently high.
Proposition 7	Flight-to-safety demand increases the incentive to sell when $s < 1/2$ and decreases it when $s > 1/2$.
Corollary 6	Flight-to-safety stabilizes already-stable markets but destabilizes fragile markets.

6 Short summary

Eisenbach and Phelan build a theory of why safe-asset markets can be fragile. The safe asset is fundamentally safe, but selling pressure must be absorbed by dealers whose balance sheets are costly. Liquidity investors hold safe assets for cash-like protection. When they fear that many others will sell and leave dealers with large inventory, they may sell preemptively even before they need cash.

The central strategic object is the payoff gain from selling now rather than waiting. Others' sales lower today's price, which discourages selling, but they also lower the future price by loading dealers with inventory, which encourages selling. When the second force dominates, strategic complementarities arise and a dash-for-cash becomes possible.

A global-game refinement turns the multiple-equilibrium logic into a threshold equilibrium. Below the threshold, the market works normally and only investors with genuine liquidity needs sell. Above the threshold, strategic investors also sell and the price drops discontinuously. Dealer balance sheet costs lower the threshold, making the market more fragile.

The most distinctive result concerns flight-to-safety. Safety investors' demand can support the future price by reducing dealer inventory, but it also raises the current price and thus rewards selling now. In a stable market, safety demand is stabilizing. In a fragile market, it can be destabilizing and can trigger a dash-for-cash.

7 Conclusion

7.1 Core conclusion

Safe asset markets can be fragile even when the asset itself has no payoff risk. Fragility comes from the strategic behavior of liquidity investors interacting with constrained dealer intermediation.

7.2 Economic intuition

A liquidity investor asks: should I sell now or wait? If other investors sell now, dealers become loaded with inventory. If I wait and later need cash, I may face a much worse price. This makes selling now privately attractive. When enough investors reason this way, a market that is normally liquid can suddenly experience a run.

7.3 Policy takeaway

Stabilization policy should reduce the fear of bad future market liquidity. Credible future purchase facilities, appropriately designed taxes or swing-pricing-like mechanisms, and relaxation of binding dealer constraints can raise the stability threshold. Policies that only make immediate selling easier, or fail to target the binding balance sheet constraint, may not stabilize the market and can even worsen fragility.

7.4 Final takeaway

The paper's main message is that a dash-for-cash is not the opposite of flight-to-safety. In a fragile safe-asset market, the two can be connected: the very demand that normally supports safe assets can, through current prices and strategic selling incentives, help trigger a sudden collapse in market liquidity.